

Auteur: Zaky Souai
Studierichting: Informatica
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$$\begin{aligned}& \int_{-1}^0 \sinh(5x) dx \\& u = 5x \Rightarrow du = 5dx \Rightarrow dx = \frac{du}{5} \\& x = -1 \Rightarrow u = -5 \\& x = 0 \Rightarrow u = 0 \\& \int_{-1}^0 \sinh(5x) dx = \int_{-5}^0 \sinh(u) \frac{du}{5} \\& = \frac{1}{5} \int_{-5}^0 \sinh(u) du \\& = \frac{1}{5} [\cosh(u)]_{-5}^0 \\& = \frac{1}{5} (\cosh(0) - \cosh(-5)) \\& = \frac{1}{5} (1 - \cosh(5)) \\& = \frac{1 - \cosh(5)}{5}\end{aligned}$$

References